

Response to the Phase 0 DB & CDN Action Plan

LocalAI TV · Architecture Verdict · B-1 / B-2 / B-3 review

To: Sheikh Sameer & Gnana Rajnan (AI Pipeline Engineers) · cc Nagarjuna Reddy (CTO), Founder Koneti Mohan Reddy **From:** LocalAI TV Architecture Team **Date:** 18 May 2026 **Subject:** Verdict on the Phase 0 DB & CDN Action Plan (17 May) — B-2 & B-3 endorsed as-is; B-1 endorsed in direction with **1 critical production-risk to resolve before executing**; still-open items confirmed **Status:** 🟡 Gate remains held. Plan is high quality. One DNS risk must be handled before B-1 execution; the previously-open items still gate the lift.

TL;DR (plain language for the Founder)

This is a strong, careful plan. Two of the three blockers are **fully and correctly solved**:

- **B-3 (database version): SOLVED.** They chose **PostgreSQL 16** (mature, proven, long support) and correctly dropped the risky PG18. Good, conservative engineering. Nothing more needed.
- **B-2 (database on the wrong cloud): SOLVED.** They will move the AI pipeline database off Google Cloud onto **AWS (same region as everything else)**. The migration plan is textbook-correct, with a 7-day fallback. Endorsed.
- **B-1 (the media web address): right direction, but one real danger.** Their plan includes an option that would **hand the entire `localaitv.com` domain's DNS control to Cloudflare**. If done without care, this can **break the API your live Play Store app already uses (`aiservices.localaitv.com`)** — i.e., the app stops working for real users. They underweighted this. It must be handled before they execute (details in §3). I recommend the safer path.

This does not lift the gate yet. It also does not change the still-open items (Admin DB verification, IAM scope, bucket, GPU approval, the C1–C4 questions, and Gnana/Abishek sign-offs). Net: the path is now much clearer; one risk to defuse + the open items to close, then re-verify, then you lift the gate.

1. B-3 — PostgreSQL 16 · ✅ Endorsed as-is

Correct decision, well-reasoned:

- **PG16 on AWS RDS** — mature, long support lifecycle, ~1 year+ production mileage, ORM/tooling well-established.
- **PG18 dropped** — exactly right; it was not GA on RDS and extension support was unverified (this was the core of my B-3 flag).
- **PG17 not adopted** — correct; none of its headline features matter at Phase 0 scale.
- **pg_cron confirmed on RDS PG16 (1.6.x), pgvector available** — verified and adequate.

Nothing further required. This blocker is closed at the decision level (execution = the RDS provisioning in B-2).

2. B-2 — DB cloud reconciliation · Endorsed as-is

This fully resolves the cross-cloud blocker I raised:

- **Migrate `newsai_db` GCP Cloud SQL → AWS RDS PostgreSQL 16, ap-south-1**, co-located with EC2 + S3. Removes the inter-cloud latency + egress + split-ops problem. Consistent with the federation model and the team's own "don't mix clouds" position.
- **Migration plan (§2.5)** — disciplined and correct: private security group (no `0.0.0.0/0` on 5432), `pg_dump` → `pg_restore` with row-count + checksum verification, smoke test, **GCP kept read-only for 7 days as fallback** then decommission with a final snapshot. Endorsed.
- **pg_cron enablement (§2.6)** — the custom parameter group / `shared_preload_libraries` / `cron.database_name` / reboot sequence is accurate and is exactly the non-obvious detail that must be in the runbook. Good.

Two execution notes (not objections — confirm in the completion report):

1. **Cutover must not disrupt the 9 live channels.** The pipeline currently runs against GCP Cloud SQL serving live channels. Run the dump→restore→repoint as a **defined low-traffic maintenance window with a write-freeze**, and a **tested rollback** (repoint `.env` back to GCP if RDS smoke-test fails). The 7-day read-only fallback supports this — confirm the live channels are verified working *after* cutover.
2. **Admin DB open item (§2.7) — correctly handled.** You rightly did **not** touch the Admin DB and flagged that it isn't in this codebase's `.env`. **This must be positively confirmed before scope freeze:** inspect the Admin Dashboard repo and report (a) is it still Supabase Mumbai? (b) if not, where? Until confirmed, "Supabase Mumbai" is an assumption, not a verified fact. This stays a gate-blocker.

3. B-1 — CDN setup · Direction endorsed · ONE critical risk to resolve first

The overall approach (create the media subdomain → Cloudflare → S3, with a pre-flight DNS check and a `cdn./media.` fallback) is correct, and the pre-flight verification (§3.2) is good discipline. **But one decision in the plan is higher-risk than presented and must be resolved before execution:**

 **B-1-CRITICAL — Do NOT do full nameserver delegation of `localaitv.com` without a DNS audit first**

§3.4 recommends **Option A: full nameserver delegation** — moving authoritative DNS for the **entire `localaitv.com` domain** to Cloudflare. The plan frames this as "easier... recommended for new projects." **`localaitv.com` is not a new project — it has live production services on it**, most importantly:

- **`aiservices.localaitv.com`** — the API base URL the **already-live Play Store app** calls (`API_BASE = https://aiservices.localaitv.com/api`).
- Plus the web app, the marketing presence, and any email/MX records.

If nameservers are switched to Cloudflare and **every existing DNS record is not first exported from Hostinger and re-created in Cloudflare**, those records stop resolving — **the live app breaks for all current users**. This is a severe, production-wide risk for what is supposed to be an isolated Phase 0 task. The plan does not mention auditing/migrating existing records.

Required before any B-1 execution — choose one:

- **(Recommended) Option C — AWS CloudFront instead of Cloudflare for the S3 media CDN.** The stack is already all-AWS in ap-south-1 (EC2 + S3 + now RDS). CloudFront in front of the S3 bucket, with a **single CNAME** `content.localaitv.com → CloudFront` added at Hostinger, leaves Hostinger as authoritative DNS, touches **nothing else** on the live domain, and keeps the media path all-AWS — the *same logic* you just used in B-2 to move the DB onto AWS to avoid mixing clouds. Lowest risk, most coherent. v1.3 specifies Cloudflare; switching the Phase-0 media CDN to CloudFront is a reasonable refinement — record it in Addendum A if adopted.
- **OR Option B — subdomain-only via Cloudflare** (keep Hostinger authoritative; do not migrate nameservers). Note the plan's caveat that partial/CNAME setup needs a paid Cloudflare plan — budget for it if this path is chosen.
- **OR Option A only with a mandatory pre-step:** export **all** current Hostinger DNS records → recreate **every** record in Cloudflare (verify `aiservices.localaitv.com`, the app, web, email resolve) → switch nameservers → re-verify the **live app still works end-to-end** → keep a tested rollback (switch nameservers back to Hostinger). Do not switch nameservers until the live-app check passes on the Cloudflare config.

Pick the DNS approach and confirm the live-app-safety plan in the completion report before executing B-1. This is the one item in the document that could cause a production outage.

B-1 minor notes (acceptable, with conditions)

- **§3.6 S3 public-read on `ai-processed/*` for Phase 0** — acceptable as a *time-boxed* Phase 0 testing measure **only if**: (a) the public-read policy is scoped to *exactly* the `ai-processed/*` prefix and nothing else (never bucket-wide), and (b) it is recorded in **Addendum A** as a mandatory Phase-1 lockdown (move to private origin + Origin Access before any production traffic). Prefer Origin Access now if it doesn't add material time.
- **§3.7 G3 acceptance criteria** — good and complete (Cloudflare/CloudFront cache header proof + repeat-request cache HIT). Keep as the G3 definition of done, with the hostname adjusted to whatever DNS path is chosen.

4. Still-Open Items — confirmed as gate-blockers (the plan correctly lists these)

None of these are closed by this document; all must close before the gate lifts:

Item	Status
Admin DB = Supabase Mumbai?	☐ Must verify from the Admin repo (B-2 §2.7)
S-1 — IAM scoped to the Phase 0 bucket + EC2-launch permission	☐ Open
S-2 — confirm <code>localaitv-content-mumbai</code> exists in ap-south-1	☐ Open
GPU quota (AWS Case 177892985400464)	☐ Confirm approved before EC2 launch
C1–C4 architecture clarifications + Gnana & Abishek sign-offs	☐ Open

5. Next Steps

1. **AI team:** pick the B-1 DNS approach and document the **live-app-safety plan** (the §3 critical item) — recommended: AWS CloudFront (Option C).
2. **AI team:** execute B-2 (RDS PG16 migration) with the maintenance-window + rollback + live-channel verification; execute B-1 on the chosen safe DNS path; verify G3.
3. **AI team:** close the §4 still-open items, including positively confirming the Admin DB host, and answer C1–C4 + collect Gnana/Abishek sign-offs.
4. **AI team → Architecture Team:** submit the completion note for **re-verification** (including the B-1 DNS-safety confirmation and post-cutover live-app/channel check).
5. **Founder:** lift the Phase 0 gate **only after** re-verification passes. Then Phase 0 execution proceeds per PHASE-0-RUNBOOK Part B.

Net: B-2 and B-3 are solved. B-1 is sound in direction but must not be executed until the DNS production-risk is defused (recommend CloudFront). The gate stays held until §4 closes and re-verification passes.

Sign-off

Party	Action	Status
Sameer & Gnana	Resolve B-1 DNS-safety (recommend CloudFront); execute B-2/B-1; close §4 items	☐ Pending
Nagarjuna (CTO)	Technical sign-off — confirm DNS approach protects the live app	☐ Pending
Architecture Team	Re-verify on completion note (DB cutover + DNS safety + G3 + §4)	☐ Awaiting
Founder — Koneti Mohan Reddy	Lift gate only after re-verification	☒ Holding

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Gate held. B-2/B-3 endorsed; B-1 direction endorsed pending the DNS production-risk fix. v1.3 unchanged; CDN-host/CloudFront substitution (if adopted) recorded in Addendum A. No Phase 0 execution until the Founder lifts the gate.